# Outline Solutions (Key Steps Only)

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1. **Solution**

$$\int_C \frac{z}{z^2+4}\,dz, \quad |z|=3$$

The singularities are at $z=\pm 2i$, both inside the contour.

$$\frac{z}{z^2+4} = \frac{1}{2}\left(\frac{1}{z-2i}+\frac{1}{z+2i}\right)$$

Sum of residues $= 1$.

$$\boxed{\int_C \frac{z}{z^2+4}\,dz = 2\pi i}$$

2. **Solution**

Let $z = e^{it}$, $dz = ie^{it}dt$, $0 \le t \le 2\pi$.

$$\Re(z) = \cos t$$

$$\oint_{|z|=1} \Re(z)\,dz = i\int_0^{2\pi} \cos t(\cos t + i\sin t)\,dt$$

Both integrals vanish over $[0, 2\pi]$.

$$\boxed{0}$$

3. **Solution**

$$2z^2 - 5z + 2 = (2z-1)(z-2)$$

Poles: $z = \frac{1}{2}$ (inside), $z = 2$ (outside).

$$\text{Res}_{z=\frac{1}{2}} = \lim_{z\to\frac{1}{2}} \frac{z-\frac{1}{2}}{(2z-1)(z-2)} = -\frac{1}{3}$$

$$\boxed{\oint_{|z|=1} \frac{1}{2z^2-5z+2}\,dz = -\frac{2\pi i}{3}}$$

4. **Solution**

$$\begin{cases} X' = 2X + Y \\ Y' = -X + 2Y \end{cases}$$

Eigenvalues: $\lambda = 2 \pm i$.

$$\boxed{\begin{aligned} X(t) &= e^{2t}(\cos t + \sin t) \\ Y(t) &= -e^{2t} \sin t \end{aligned}}$$

5. **Solution** NEED

6. **Solution**

$$\mathcal{L}\{\sin(3t)\} = \int_0^\infty e^{-st} \sin(3t)\, dt$$

$$\boxed{\mathcal{L}\{\sin(3t)\} = \frac{3}{s^2 + 9}}$$

7. **Solution**

$$\mathcal{L}^{-1}\left\{ \frac{s^2 + 2s + 3}{(s^2 + 2s + 2)(s^2 + 2s + 5)} \right\}$$

$$= \frac{1}{3}\left( \frac{1}{s^2 + 2s + 2} + \frac{2}{s^2 + 2s + 5} \right)$$

Using the shift theorem,

$$\boxed{f(t) = \frac{1}{3} e^{-t} (\sin t + 2 \sin 2t)}$$